Aadi Ramnani

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Education

CHIREC International School *Hyderabad, India*
IB Diploma Programme – Year 2 | Predicted 7 in 5 subjects, 6 in English *Aug 2025 – May 2027*

- HL: Physics 7 · Chemistry 7 · Mathematics AA 7 SL: Business 7 · English 6 · French 7

Cambridge IGCSE | 8 A*s, 1 A · Avg PUM 94 · ICE with Distinction · World Topper – Chemistry *Completed May 2025*

- Chemistry 100 · Biology 97 · Mathematics 97 · Physics 96 · CS 95 · Add. Math 95 · Economics 94 · French 90 · English 84

Standardised Testing: SAT 1540 (Math 790 / EBRW 750) *June 2025*

Technical Skills

- **Programming:** Python (NumPy, Matplotlib), TypeScript, HTML5, CSS3
- **Scientific Computing:** First-principles / DFT workflows (structural relaxation, electronic density of states, zone-centre phonon frequencies); lattice dynamics
- **Machine Learning:** Supervised Fine-Tuning, Group Relative Policy Optimization (GRPO), reward shaping, model quantization & edge deployment
- **CFD & Simulation:** SimScale, OpenFOAM, SimFlow, BaramFlow, FluidX3D – flow visualisation, pressure distribution, boundary-layer separation
- **Engineering & CAD:** Autodesk Fusion 360, FDM 3-D printing (material selection, print optimisation), rapid prototyping, iterative design cycles

Experience

Research Intern – Computational Physics *Apr 2026 – Present*
Computational Physics Group, Prof. Manish K. Niranjan, IIT Hyderabad *15–20 hrs/wk (summer); 8–10 hrs/wk (term)*

- Ran first-principles quantum-mechanical simulations of the $LaAlO_3/SrTiO_3$ oxide interface, where two bulk insulators meet to host a conducting 2-D electron gas; ran structural relaxations, density-of-states and zone-centre phonon calculations on thin superlattice models.
- Identified vibrational modes localised at the interface and reasoned that Raman spectroscopy could detect them; wrote Python scripts to process and analyse the simulation output.
- Co-author on a research manuscript in preparation for submission to a peer-reviewed journal.

Machine Learning Intern *Summer 2025*
OrcaLex Technologies – SmartLearners.ai, Hyderabad *Production AI platform: ~20,000 students, 600 teachers*

- Improved the accuracy with which a production vision-language model reads handwritten exam notation while cutting inference cost (token-generation speed on NVIDIA A100 GPUs), as an experimentalist – isolating one variable at a time (training method, data composition, decoding parameters) against teacher-marked ground truth.
- Tuned Supervised Fine-Tuning and GRPO reward functions – penalising needlessly long reasoning chains to produce denser, faster, cheaper outputs without losing marking accuracy; tracked character-level error, symbol-recovery rates and memory footprints across configurations.
- Carried the best checkpoints through to quantized edge-deployment formats; on finding an analysis script had double-counted students, flagged and corrected it unprompted and re-ran the full analysis.

Chief Aerodynamicist & Design Engineer *Sept 2025 – Present*
Ashwa Racing – STEM Racing / F1 in Schools, CO_2-Powered Vehicle | Design Process Doc

- Solo-designed the team's car across 6 full prototype cycles and 50+ component iterations, grounding each in aerodynamic theory and validating it through CFD in 3+ solvers – the design that reached Nationals.
- Placed **2nd Runner-Up among 100+ teams** at the South India Regionals (team's highest-ever placement) and **6th at Nationals**.
- Used first-principles force, energy and momentum analysis to show wheel traction raises rolling losses rather than forward force, correcting a core team misconception; CFD-compared nosecone geometries (elliptical, ogive, von Kármán/Haack) and rejected intuitive-but-inefficient concepts (rifling, Magnus effect, excess downforce).

Co-Founder & Product Lead – TutorLink *6-month development cycle*
Peer-to-Peer Academic Platform (Ed-Tech Startup) | tutorlink.world

- Co-founded a peer-tutoring platform with transparent, blockchain-logged contribution tracking (Polygon); ran 2-stage market research with 125+ respondents (81% stated adoption); ML tutor–learner matching at 0.86 F1.
- Pitched to industry judges – **Runner-Up** at a competitive venture showcase.

Research & Projects

Project AKS – Hypergolic Rocket System Feasibility Study *Team Lead | Independent Research*

- Led a systems-level feasibility study for a reusable UDMH–N_2O_4 rocket deployed via high-altitude balloon for near-space access (80–100 km); modelled injector/nozzle configurations, thrust profiles and mass–altitude trade-offs (Tsiolkovsky equation); evaluated regenerative and ablative thermal management under safety constraints.
- Presented the full technical proposal to school leadership; project halted after permits review; preparing it as a publicly citable preprint.

Chemistar – AI-Powered Virtual Chemistry Laboratory *TypeScript, NLP, Simulation*

- Built an AI-driven web platform for free-form virtual experimentation with unsafe, rare or cost-prohibitive reactions (250+ reaction types); NLP parser for natural-language experiment design at 0.82 F1; dynamic engine models real-time temperature and pH without pre-scripted reactions.
- **1st place, C-HACKS**; presented to 150+ attendees including Microsoft leadership; coaches IGCSE Chemistry and built resources used by 3 teachers.

Air & Water Rocket – Iterative Prototyping *Design Iteration Report*

- Built 6 prototypes and 12 launch-pad architectures through failure-driven iteration; tuned air-to-water ratio to 1:3 and pressure thresholds; integrated altimeter, pressure and GPS sensors to validate against simulations; converged on a carbon-fibre final configuration.

IBDP Collaborative Sciences Project – Street-Food Safety Study *Team Lead | Slide Deck*

- Led a multidisciplinary team across chemistry, biology and physics; personally ran heat-transfer modelling and thermodynamic analysis of real-world cooking and storage. **1st Place** – "Outstanding Exploration and Integration of a Significant Global Scientific Issue."

Leadership & Activities

President & Co-Founder – The Drama Collective *Ongoing*

Founder-Led Theatre Initiative – 50+ members across ~10 departments | Website

- Revived theatre after the school had largely discontinued it: won approval across leadership meetings, then ran the Collective solo after both co-founders left – now 50+ members across ~10 departments (directing, scriptwriting, publication, outreach, costume, set, sound), each with its own lead and sub-team; fully self-funded.
- Spearheading the school's first large-scale, fully student-led production, alongside (to my knowledge) India's first student-led theatre-centric publication and cross-school outreach including a "mute theatre" collaboration with Project SignVerse; built nepotism-resistant audition councils and department charters so it outlasts any cohort.

Theatre & Performance *Since Kindergarten*

- Lead roles since 2012 (Helen O'Grady, 2012–2015; The International Drama Academy, 2017–2020); danced in three Shiamak Davar stage shows (2017–2019).
- Wrote and directed original social plays on teen alcohol abuse, over-reliance on AI, cultural barriers and child safety; played both male and female roles.

Cyberbullying Lead – School-wide Safeguarding Initiative *12+ months*

- Coordinated 97 student ambassadors and 11 faculty mentors; designed and delivered 20 workshops reaching ~1,000 students; realigned 6 institutional policies to the Diana Award Anti-Bullying Ambassador framework; launched a weekly advocacy newsletter and parent sessions.

Co-Founder & Instructor – 3D Printing Club *Grade 8 (2022–2023)*

- Youngest student to launch an after-school technical club at CHIREC; taught CAD (Fusion 360, Cura) and additive manufacturing; members later independently built the first physical prototype, demonstrating effective knowledge transfer.

Sustainability & Community Service *3+ years*

- **Litter League (school waste-management club):** designed waste-segregation workflows and ran segregation at major school events (incl. CHIREC MUN), achieving ~40% food-waste reduction in week 1.
- **Zero-Waste Outreach (independent):** designed and delivered 4 awareness sessions on zero-waste living and recycling to 500+ students (Grades 2–8) and a local orphanage; joined lake clean-ups across Hyderabad.
- **Project SignVerse:** completed months of sign-language training; 1 of ~7 students chosen for the Sign-A-Thon accessibility walk. **Interact Club:** 3 years of fundraising and outreach for children living with HIV/AIDS.

Awards & Certifications

- **Outstanding Cambridge Learner Award – Top in the World (Chemistry)**, Cambridge Assessment International Education, 2025
- **Bronze Award** – Singapore and Asian Schools Math Olympiad (SASMO), 2025
- **Selective Summer Programs:** Summer Science Program (SSP) · VTSP
- **Certifications:** Data Science & AI – IIT Madras (2026) · Quantum Computing – IIT Hyderabad (2025) · Software Development – IIT Hyderabad (2025) · Premier Aerodynamics – F1 in Schools (2025) · Technology & Sustainability – IBM (2023) · Industry & Innovation – T-HUB (2024)
- **IBM / Coursera (2025):** Software Engineering · Intro to AI · Generative AI · Prompt Engineering · HTML/CSS/JS
- **Hackathons:** IIT Jodhpur – Flying Wings · Mission Brain